

Tutorial 3

1. The longitudinal quartic for a mid-sized transport aircraft is

$$\lambda^4 + 2.00\lambda^3 + 3.01\lambda^2 + x\lambda + 0.03 = 0.$$

- If the aircraft's phugoid mode is neutrally (un)stable, find the value of x , which is small.
- Hence find the period and damping of all the longitudinal natural modes of the aircraft.
- Based on your findings, what is the cruising speed of the aircraft?

a. if phugoid mode is neutrally stable, then $\lambda = 0 \pm ai = \pm ai$

$$\therefore (\pm ai)^4 + 0.03(\pm ai)^3 + 3.01(\pm ai)^2 \pm xai + 0.03 = 0$$

$$\therefore a^4 \mp 2a^3i - 3.01a^2 \pm xai + 0.03 = 0$$

$$\therefore (a^4 - 3.01a^2 + 0.03)[\pm(-2a^3 + xa)i] = 0$$

$$\text{for } a^4 - 3.01a^2 + 0.03 = 0, \quad a^2 = \frac{3.01 \pm \sqrt{8.9401}}{2} = 3 \text{ or } 0.01$$

$\therefore a^2 = 0.01$ for phugoid mode which is relatively small

$$\therefore a = \pm 0.1$$

$$\text{for } 2a^3 - xa = 0, \quad 0.002 - 0.1x = 0$$

$$\therefore x = 0.02$$

(a) By Routh's discriminant, $R = 0$ for neutral stable

$$R = BCD - B^2E - AD^2 = 6.02x - 0.12 - x^2 = 0$$

$$\therefore x = \frac{-6.02 \pm \sqrt{35.7604}}{-2} = 6 \text{ or } 0.02$$

$$\therefore x = 0.02 \text{ for phugoid mode} \quad \therefore a = \pm 0.1i$$

(b) for phugoid mode, $a = \pm 0.1i$ $\therefore \lambda = \pm 0.1i$ $\therefore \omega_d = \omega_n = 0.1$

$$\therefore T = \frac{2\pi}{\omega_d} = \frac{2\pi}{0.1} = 20\pi$$

$$\therefore \zeta = \frac{|c|}{|\omega_n|} = \frac{0}{0.1} = 0$$

for SPPD mode, $(A\lambda^2 + B\lambda + C)(\lambda - 0.1i)(\lambda + 0.1i) = 0$

$$\therefore (A\lambda^2 + B\lambda + C)(\lambda^2 + 0.01) = 0$$

$$\therefore A\lambda^4 + B\lambda^3 + (C + 0.01A)\lambda^2 + 0.01B\lambda + 0.01C = 0$$

$$\therefore A = 1, B = 2, C = 3$$

$$\therefore \text{for } A\lambda^2 + B\lambda + C, \lambda = \frac{-2 \pm \sqrt{4 - 12}}{2} = -1 \pm \sqrt{2}i$$

$$\therefore \omega_d = \sqrt{2} \quad \therefore T = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$$

$$\therefore \zeta = \frac{|c|}{|\omega_n|} = \frac{1}{\sqrt{1^2 + \sqrt{2}^2}} = \frac{1}{3}$$

c. for phugoid mode, $\omega_d = \sqrt{2} \frac{g}{U_e} = 0.1$

$$\therefore U_e = \frac{\sqrt{2}g}{0.1} = 130.73 \text{ m/s}$$

2. The lateral stability quintic of a large airliner is:

$$\lambda^5 + a\lambda^4 + 1.1025\lambda^3 + 1.0025\lambda^2 = 0.$$

The aircraft is known to have a stable roll subsidence mode with a time to half amplitude of 0.6931 s.

- (a) Approximate the value of the unknown parameter a .
- (b) Will this aircraft be dynamically stable in the spiral mode? What are the physical implications of your findings?
- (c) What is the frequency of oscillation and damping ratio of this aircraft's Dutch Roll mode.
- (d) What behaviour does the lateral eigenvalue $\lambda = 0$ that aircraft typically exhibit correspond to?

$$(a) \quad t_{\frac{1}{2}} = \frac{\ln 2}{|r_j|} \quad \therefore \lambda = r_j \text{ for roll subsidence} = -1$$

$$\therefore (-1)^5 + a(-1)^4 + 1.1025(-1)^3 + 1.0025(-1)^2 = 0$$

$$\therefore -1 + a - 1.1025 + 1.0025 = a - 1.1 = 0$$

$$\therefore a = 1.1$$

$$(b) \quad \lambda^5 + 1.1\lambda^4 + 1.1025\lambda^3 + 1.0025\lambda^2 = 0$$

$$\lambda^2(\lambda^3 + 1.1\lambda^2 + 1.1025\lambda + 1.0025) = 0$$

$\therefore$ There are two neutrally stable mode

$\therefore$ Spiral mode would be neutrally stable

$\therefore$ Aircraft will turn at constant angle once input a turn

$$(c) \quad \lambda^3 + 1.1\lambda^2 + 1.1025\lambda + 1.0025 = (\lambda + 1)(\lambda^2 + B\lambda + C)$$

$$= \lambda^3 + (B+1)\lambda^2 + (B+C)\lambda + C = 0$$

$$\therefore B = 0.1, C = 1.0025$$

$$\therefore \text{for Dutch roll, } \lambda^2 + 0.1\lambda + 1.0025 = 0$$

$$\therefore \lambda = \frac{-0.1 \pm \sqrt{-4}}{2} = -0.05 \pm i$$

$$\therefore \omega_d = 1 \quad \therefore T = \frac{2\pi}{\omega_d} = 2\pi$$

$$\therefore \xi_1 = \frac{0.05}{\sqrt{0.05^2 + 1}} = 0.049938$$

d) Aircraft is neutrally stable with respect to a change in the defination of its heading angle

3. The dynamics of a rotorcraft can be represented in state space by the system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$

$$\begin{bmatrix} \dot{q} \\ \dot{\theta} \\ \dot{U} \end{bmatrix} = \begin{bmatrix} -0.4 & 0 & -0.01 \\ 1 & 0 & 0 \\ -1.4 & 9.8 & -0.02 \end{bmatrix} \begin{bmatrix} q \\ \theta \\ U \end{bmatrix} + \begin{bmatrix} 6.3 \\ 0 \\ 9.8 \end{bmatrix} u(t)$$

where $u(t)$ is the control input being applied to the system.

- Is the open-loop system stable?
- Derive the transfer functions, in frequency domain, for the states of the system with respect to the control input.
- Hence calculate the effect of a step control input on the rotorcraft's velocity U
- If a feedback control system is implemented such that the control input

$$u(t) = -[0.4706 \quad 1 \quad 0.0627] \mathbf{x}(t),$$

show that the system is reduced to the following representation. What type of controller is that?

$$\begin{bmatrix} \dot{q} \\ \dot{\theta} \\ \dot{U} \end{bmatrix} = \begin{bmatrix} -3.3648 & -6.3000 & -0.4050 \\ 1 & 0 & 0 \\ -6.0119 & 0 & -0.6345 \end{bmatrix} \begin{bmatrix} q \\ \theta \\ U \end{bmatrix}$$

- For the closed-loop system given in (d), calculate the closed-loop response of the system from an initial condition $[1, 1, 1]$.

a. for open loop system,

$$\begin{vmatrix} -0.4 - \lambda & 0 & -0.01 \\ 1 & -\lambda & 0 \\ -1.4 & 9.8 & -0.02 - \lambda \end{vmatrix}$$

$$\therefore -1 \cdot 9.8 \cdot 0.01 - \lambda [(-0.4 - \lambda)(-0.02 - \lambda) - (-0.01)(-1.4)] = 0$$

$$\therefore -\lambda^3 - 0.42\lambda^2 + 0.006\lambda - 0.098 = 0$$

$$\therefore \lambda_1 = -0.65662, \quad \lambda_{2,3} = 0.1178 \pm 0.3682i$$

$\therefore$ unstable

OR

for Routh's Discriminant, $A\lambda^3 + B\lambda^2 + C\lambda + D = 0$

$A, B, C, D > 0$ and $AD > BC$ for stable

$$b) sX(s) = AX(s) + BU(s)$$

$$\therefore (Is - A)X(s) = BU(s)$$

$$\therefore G(s) = \frac{X(s)}{U(s)} = \frac{B}{Is - A} = (Is - A)^{-1} B$$

$$Is - A = \begin{bmatrix} s + 0.4 & 0 & 0.01 \\ -1 & s & 0 \\ 1.4 & -9.8 & s + 0.02 \end{bmatrix}$$

$$\therefore (Is - A)^{-1} = \frac{\text{adj}(Is - A)}{\det(Is - A)} = \frac{C^T}{\det(Is - A)}$$

$$= \frac{\begin{bmatrix} s^2 + 0.02s & s + 0.02 & -1.4s + 9.8 \\ -0.098 & s^2 + 0.42s - 0.006 & 9.8s + 3.92 \\ -0.01s & -0.01 & s^2 + 0.4s \end{bmatrix}^T}{s^3 + 0.42s^2 - 0.006s + 0.098}$$

$$= \frac{\begin{bmatrix} s^2 + 0.02s & -0.098 & -0.01s \\ s + 0.02 & s^2 + 0.42s - 0.006 & -0.01 \\ -1.4s + 9.8 & 9.8s + 3.92 & s^2 + 0.4s \end{bmatrix}}{s^3 + 0.42s^2 - 0.006s + 0.098}$$

$$\therefore G(s) = \frac{\begin{bmatrix} s^2 + 0.02s & -0.098 & -0.01s \\ s + 0.02 & s^2 + 0.42s - 0.006 & -0.01 \\ -1.4s + 9.8 & 9.8s + 3.92 & s^2 + 0.4s \end{bmatrix} \begin{bmatrix} 6.3 \\ 0 \\ 9.8 \end{bmatrix}}{s^3 + 0.42s^2 - 0.006s + 0.098}$$

$$= \frac{\begin{bmatrix} 6.3s^2 + 0.028s \\ 6.3s + 0.028 \\ 9.8s^2 - 4.9s + 61.74 \end{bmatrix}}{s^3 + 0.42s^2 - 0.006s + 0.098}$$

c) $G(s)$ for $U = \frac{9.8s^2 - 4.9s + 61.74}{s^3 + 0.42s^2 - 0.006s + 0.098} = \frac{U(s)}{u(s)}$

where $u(s) = \frac{1}{s} u^*$

$$\therefore \frac{U(s)}{u^*} = \frac{9.8s^2 - 4.9s + 61.74}{s(s + 0.65662)(s^2 - 0.2366s + 0.1493)}$$

... see solutions

d) P controller. where $u = Cx$

$$\therefore e = r - u = -Cx$$

$$\therefore \dot{x} = Ax + Bu = (A - BC)x$$

$$\therefore \begin{bmatrix} \dot{q} \\ \dot{\theta} \\ \dot{U} \end{bmatrix} = \left(\begin{bmatrix} -0.4 & 0 & -0.01 \\ 1 & 0 & 0 \\ -1.4 & 9.8 & -0.02 \end{bmatrix} - \begin{bmatrix} 6.3 \\ 0 \\ 9.8 \end{bmatrix} [0.4706 \ 1 \ 0.0627] \right) \begin{bmatrix} q \\ \theta \\ U \end{bmatrix}$$

$$\therefore \begin{bmatrix} \dot{q} \\ \dot{\theta} \\ \dot{U} \end{bmatrix} = \left(\begin{bmatrix} -0.4 & 0 & -0.01 \\ 1 & 0 & 0 \\ -1.4 & 9.8 & -0.02 \end{bmatrix} - \begin{bmatrix} 2.96478 & 6.3 & 0.39501 \\ 0 & 0 & 0 \\ 4.61188 & 9.8 & 0.61446 \end{bmatrix} \right) \begin{bmatrix} q \\ \theta \\ U \end{bmatrix}$$

$$= \begin{bmatrix} -3.36478 & -6.3 & -0.40501 \\ 1 & 0 & 0 \\ -6.01188 & 0 & -0.63446 \end{bmatrix} \begin{bmatrix} q \\ \theta \\ U \end{bmatrix}$$

(e) detail see solution

4. The longitudinal motion of an aerial vehicle is given by

$$\begin{aligned}\dot{w} &= -2w + 197\dot{\theta} - 27\delta \\ \ddot{\theta} &= -0.25w - 15\dot{\theta} - 45\delta\end{aligned}$$

- (a) Pick an appropriate set of states and express the dynamics of the system in state-space form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u.$$

- (b) Determine the stability of the open-loop system.

- (c) Determine the proportional state feedback control law

$$u = \mathbf{K}\mathbf{x}$$

required such that the closed-loop system response is oscillatory with a damping ratio $\zeta = 0.5$ and an undamped natural frequency $\omega_n = 20$ rad/s.

$$(a) \begin{bmatrix} \dot{w} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} -2 & 197 \\ -0.25 & -15 \end{bmatrix} \begin{bmatrix} w \\ \theta \end{bmatrix} + \begin{bmatrix} -27 \\ -45 \end{bmatrix} \begin{bmatrix} \delta \end{bmatrix}$$

$$(b) \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$\text{Let } \mathbf{x} = \mathbf{x}^* e^{\lambda t} \quad \therefore \dot{\mathbf{x}} = \lambda \mathbf{x}^* e^{\lambda t}$$

$$\therefore (\mathbf{A} - \lambda \mathbf{I}) \mathbf{x} + \mathbf{B}u = 0$$

$$\therefore |\mathbf{A} - \lambda \mathbf{I}| = 0$$

$$\therefore \begin{vmatrix} -2 - \lambda & 197 \\ -0.25 & -15 - \lambda \end{vmatrix} = \lambda^2 + 17\lambda + 79.25 = 0$$

$$\therefore \lambda = \frac{-17 \pm \sqrt{289 - 317}}{2} = -\frac{17}{2} \pm 2\sqrt{7}i$$

$\therefore$ oscillatory stable

$$\begin{aligned} \text{'c')} \quad \begin{bmatrix} \dot{\omega} \\ \ddot{\theta} \end{bmatrix} &= \left(\begin{bmatrix} -2 & 197 \\ -0.25 & -15 \end{bmatrix} + \begin{bmatrix} -27 \\ -45 \end{bmatrix} \begin{bmatrix} K_1 & K_2 \end{bmatrix} \right) \begin{bmatrix} \omega \\ \theta \end{bmatrix} \\ &= \begin{bmatrix} -2 - 27K_1 & 197 - 27K_2 \\ -0.25 - 45K_1 & -15 - 45K_2 \end{bmatrix} \begin{bmatrix} \omega \\ \theta \end{bmatrix} \end{aligned}$$

$$\therefore \begin{vmatrix} -2 - 27K_1 - \lambda & 197 - 27K_2 \\ -0.25 - 45K_1 & -15 - 45K_2 - \lambda \end{vmatrix} = 0$$

$$\begin{aligned} \therefore [\lambda^2 + (27K_1 + 45K_2 + 17)\lambda + 1215K_1K_2 + 405K_1 + 90K_2 + 30] \\ - (1215K_1K_2 - 8865K_1 + 6.75K_2 - 49.25) = 0 \end{aligned}$$

$$\therefore \lambda^2 + (27K_1 + 45K_2 + 17)\lambda + 9270K_1 + 83.25K_2 + 79.25 = 0$$

$$\therefore \lambda = \frac{-(27K_1 + 45K_2 + 17) \pm \sqrt{(27K_1 + 45K_2 + 17)^2 - 4(9270K_1 + 83.25K_2 + 79.25)}}{2}$$

$$= \lambda \pm si$$

$$\therefore \zeta = \frac{-\lambda}{\omega_n} = \frac{-\lambda}{\sqrt{\lambda^2 + s^2}} = 0.5 \quad \text{while} \quad \omega_n = 20 \text{ rad/s}$$

$$\therefore \lambda^2 + 2\zeta\omega_n \lambda + \omega_n^2 = 0$$

$$\therefore \lambda^2 + 2 \cdot 5 \cdot 20 \lambda + 20^2 = 0$$

$$\therefore \lambda^2 + 200\lambda + 40 = 0$$

$$\therefore \begin{cases} 27K_1 + 45K_2 + 17 = 200 \\ 9270K_1 + 83.25K_2 + 79.25 = 40 \end{cases}$$

$$\therefore \begin{cases} K_1 = 0.0342 \\ K_2 = 0.0462 \end{cases}$$

Review 1

5. Consider the pendulum on a cart system you analysed in tutorial sheet 1. The unstable equilibrium of this system can be controlled by applying an external axial force F_x in response to any change in angle θ about the desired equilibrium condition.

(a) Express the equations governing the motion of the system in the state space form

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u.$$

- (b) For the mass and dimensions of the system presented in Tutorial 1, plot a root-locus plot to illustrate the effect of a proportional gain of the closed-loop dynamics of the system.

The equations of motion of a pendulum of mass m and length L , mounted on a cart of mass M , effectively a Segway, are

$$\ddot{\theta} = \frac{3}{2} \left(\frac{g \sin \theta - \ddot{x} \cos \theta}{L} \right)$$
$$\ddot{x}(M + m) + \frac{mL}{2}(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) = 0$$

(a) $\mathbf{M}\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\dot{\mathbf{x}}$

$$\therefore \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & M+m \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$= \begin{bmatrix} \dot{\theta} \\ \frac{3}{2L}(g \sin \theta - \ddot{x} \cos \theta) \\ \frac{mL}{2}(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ \frac{3}{2L}(g \cos \theta + \ddot{x} \sin \theta) & 0 & 0 \\ \frac{mL}{2}(-\ddot{\theta} \sin \theta - \dot{\theta}^2 \cos \theta) & -mL\dot{\theta} \sin \theta & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \\ \dot{x} \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -\frac{3}{2L} \cos \theta \\ -mL\dot{\theta} \sin \theta & \frac{mL}{2} \cos \theta & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$\therefore \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & \frac{3}{2L} \cos \theta \\ mL\dot{\theta} \sin \theta & -\frac{mL}{2} \cos \theta & M+m \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ \frac{3}{2L}(g \cos \theta + \ddot{x} \sin \theta) & 0 & 0 \\ \frac{mL}{2}(-\ddot{\theta} \sin \theta - \dot{\theta}^2 \cos \theta) & -mL\dot{\theta} \sin \theta & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \\ \dot{x} \end{bmatrix}$$

For $M\ddot{x} = Ax + Bu$,

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & \frac{3}{2L} \cos \theta \\ mL\dot{\theta} \sin \theta & -\frac{mL}{2} \cos \theta & M+m \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ \frac{3}{2L}(g \cos \theta + \ddot{x} \sin \theta) & 0 & 0 \\ \frac{mL}{2}(-\ddot{\theta} \sin \theta - \dot{\theta}^2 \cos \theta) & -mL\dot{\theta} \sin \theta & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \\ \dot{x} \end{bmatrix}$$

+ see solution

(b) see solution